SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Revision Date 19-Mar-2024 **Revision Number** 4

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Oxalic acid, 10% w/v solution in ethanol**
Cat No. : **S55340**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids Category 2 (H225)

Health hazards

Serious Eye Damage/Eye Irritation	Category 1 (H318)
Specific target organ toxicity - (single exposure)	Category 2 (H371)
Specific target organ toxicity - (repeated exposure)	Category 2 (H373)

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Signal Word **Danger**

Hazard Statements
H225 - Highly flammable liquid and vapor
H318 - Causes serious eye damage
H371 - May cause damage to organs
H373 - May cause damage to organs through prolonged or repeated exposure
EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P280 - Wear eye protection/ face protection
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

2.3. Other hazards

Contains a known or suspected endocrine disruptor
Contains a substance on the National Authorities Endocrine Disruptor Lists

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Ethyl alcohol	64-17-5	200-578-6	74.15	Flam. Liq. 2 (H225)

				Eye Irrit. 2 (H319)
Ethanedioic acid, dihydrate	6153-56-6		11.2	Acute Tox. 4 (H302) Acute Tox. 4 (H312) Eye Dam. 1 (H318) STOT RE 2 (H373)
Ethyl acetate	141-78-6	EEC No. 205-500-4	8.88	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) EUH066
Methanol	67-56-1	200-659-6	4.44	Flam. Liq. 2 (H225) Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 3 (H331) STOT SE 1 (H370)
Methyl ethyl ketone	78-93-3	EEC No. 201-159-0	1.33	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) (EUH066)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Ethyl alcohol	Eye Irrit. 2 :: C>=50%	-	-
Methanol	STOT Single Exp. 1 :: >= 10 STOT Single Exp. 2 :: 3 - < 10	-	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion Clean mouth with water and drink afterwards plenty of water.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Carbon dioxide (CO_2). Powder. Foam. Water may be ineffective. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)	Class 3
Switzerland - Storage of hazardous substances	Storage class - SC 3 https://www.kvu.ch/de/themen/stoffe-und-produkte https://www.kvu.ch/fr/themes/substances-et-produits https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Ethyl alcohol		TWA: 1000 ppm TWA; 1920 mg/m³ TWA WEL - STEL: 3000 ppm STEL; 5760 mg/m³ STEL	TWA / VME: 1000 ppm (8 heures). TWA / VME: 1900 mg/m³ (8 heures). STEL / VLCT: 5000 ppm. STEL / VLCT: 9500 mg/m³.	TWA: 1000 ppm 8 uren TWA: 1907 mg/m³ 8 uren	STEL / VLA-EC: 1000 ppm (15 minutos). STEL / VLA-EC: 1910 mg/m³ (15 minutos).
Ethanedioic acid, dihydrate				TWA: 1 mg/m³ 8 uren STEL: 2 mg/m³ 15 minuten	
Ethyl acetate	TWA: 734 mg/m³ (8h) TWA: 200 ppm (8h) STEL: 1468 mg/m³ (15min) STEL: 400 ppm (15min)	STEL: 1468 mg/m³ 15 min STEL: 400 ppm 15 min TWA: 734 mg/m³ 8 hr TWA: 200 ppm 8 hr	TWA / VME: 200 ppm (8 heures). TWA / VME: 734 mg/m³ (8 heures). STEL / VLCT: 400 ppm. restrictive limit STEL / VLCT: 1468 mg/m³. restrictive limit	TWA: 200 ppm 8 uren TWA: 734 mg/m³ 8 uren STEL: 400 ppm 15 minuten STEL: 1468 mg/m³ 15 minuten	STEL / VLA-EC: 400 ppm (15 minutos). STEL / VLA-EC: 1468 mg/m³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 734 mg/m³ (8 horas)
Methanol	TWA: 200 ppm 8 hr TWA: 260 mg/m³ 8 hr Skin	WEL - TWA: 200 ppm TWA; 266 mg/m³ TWA WEL - STEL: 250 ppm STEL; 333 mg/m³ STEL	TWA / VME: 200 ppm (8 heures). restrictive limit TWA / VME: 260 mg/m³ (8 heures). restrictive limit STEL / VLCT: 1000 ppm. restrictive limit STEL / VLCT: 1300 mg/m³. restrictive limit Peau	TWA: 200 ppm 8 uren TWA: 266 mg/m³ 8 uren STEL: 250 ppm 15 minuten STEL: 333 mg/m³ 15 minuten Huid	TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 266 mg/m³ (8 horas) Piel
Methyl ethyl ketone	TWA: 200 ppm (8h) TWA: 600 mg/m³ (8h) STEL: 300 ppm (15min) STEL: 900 mg/m³ (15min)	STEL: 300 ppm 15 min STEL: 899 mg/m³ 15 min TWA: 200 ppm 8 hr TWA: 600 mg/m³ 8 hr Skin	TWA / VME: 200 ppm (8 heures). restrictive limit TWA / VME: 600 mg/m³ (8 heures). restrictive limit STEL / VLCT: 300 ppm. restrictive limit STEL / VLCT: 900	TWA: 200 ppm 8 uren TWA: 600 mg/m³ 8 uren STEL: 300 ppm 15 minuten STEL: 900 mg/m³ 15 minuten	STEL / VLA-EC: 300 ppm (15 minutos). STEL / VLA-EC: 900 mg/m³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 600 mg/m³ (8 horas)

			mg/m³. restrictive limit Peau		

Component	Italy	Germany	Portugal	The Netherlands	Finland
Ethyl alcohol		200 ppm TWA MAK; 380 mg/m³ TWA MAK	STEL: 1000 ppm 15 minutos	huid STEL: 1900 mg/m³ 15 minuten TWA: 260 mg/m³ 8 uren	TWA: 1000 ppm 8 tunteina TWA: 1900 mg/m³ 8 tunteina STEL: 1300 ppm 15 minuutteina STEL: 2500 mg/m³ 15 minuutteina
Ethyl acetate	TWA: 734 mg/m³ 8 ore. Time Weighted Average TWA: 200 ppm 8 ore. Time Weighted Average STEL: 1468 mg/m³ 15 minuti. Short-term STEL: 400 ppm 15 minuti. Short-term	TWA: 200 ppm (8 Stunden). AGW - exposure factor 2 TWA: 730 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 200 ppm (8 Stunden). MAK TWA: 750 mg/m³ (8 Stunden). MAK Höhepunkt: 400 ppm Höhepunkt: 1500 mg/m³	STEL: 1468 mg/m³ 15 minutos STEL: 400 ppm 15 minutos TWA: 200 ppm 8 horas TWA: 734 mg/m³ 8 horas	STEL: 1468 mg/m³ 15 minuten TWA: 734 mg/m³ 8 uren	TWA: 200 ppm 8 tunteina TWA: 730 mg/m³ 8 tunteina STEL: 400 ppm 15 minuutteina STEL: 1470 mg/m³ 15 minuutteina
Methanol	TWA: 200 ppm 8 ore. Time Weighted Average TWA: 260 mg/m³ 8 ore. Time Weighted Average Pelle	100 ppm TWA MAK; 130 mg/m³ TWA MAKSkin absorber	STEL: 250 ppm 15 minutos TWA: 200 ppm 8 horas TWA: 260 mg/m³ 8 horas Pele	huid TWA: 133 mg/m³ 8 uren	TWA: 200 ppm 8 tunteina TWA: 270 mg/m³ 8 tunteina STEL: 250 ppm 15 minuutteina STEL: 330 mg/m³ 15 minuutteina Iho
Methyl ethyl ketone	TWA: 200 ppm 8 ore. Time Weighted Average TWA: 600 mg/m³ 8 ore. Time Weighted Average STEL: 300 ppm 15 minuti. Short-term STEL: 900 mg/m³ 15 minuti. Short-term	TWA: 200 ppm (8 Stunden). AGW - exposure factor 1 TWA: 600 mg/m³ (8 Stunden). AGW - exposure factor 1 TWA: 200 ppm (8 Stunden). MAK TWA: 600 mg/m³ (8 Stunden). MAK Höhepunkt: 200 ppm Höhepunkt: 600 mg/m³ Haut	STEL: 300 ppm 15 minutos STEL: 900 mg/m³ 15 minutos TWA: 200 ppm 8 horas TWA: 600 mg/m³ 8 horas	huid STEL: 900 mg/m³ 15 minuten TWA: 590 mg/m³ 8 uren	TWA: 20 ppm 8 tunteina TWA: 60 mg/m³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 300 mg/m³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Ethyl alcohol	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 3800 mg/m³ 15 Minuten MAK-TMW: 1000 ppm 8 Stunden MAK-TMW: 1900 mg/m³ 8 Stunden	TWA: 1000 ppm 8 timer TWA: 1900 mg/m³ 8 timer STEL: 2000 ppm 15 minutter STEL: 3800 mg/m³ 15 minutter	STEL: 1000 ppm 15 Minuten STEL: 1920 mg/m³ 15 Minuten TWA: 500 ppm 8 Stunden TWA: 960 mg/m³ 8 Stunden	TWA: 1900 mg/m³ 8 godzinach	TWA: 500 ppm 8 timer TWA: 950 mg/m³ 8 timer STEL: 625 ppm 15 minutter. value calculated STEL: 1187.5 mg/m³ 15 minutter. value calculated
Ethyl acetate	MAK-KZGW: 400 ppm 15 Minuten MAK-KZGW: 1468 mg/m³ 15 Minuten MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 734 mg/m³ 8 Stunden	TWA: 150 ppm 8 timer TWA: 540 mg/m³ 8 timer STEL: 1468 mg/m³ 15 minutter STEL: 400 ppm 15 minutter	STEL: 400 ppm 15 Minuten STEL: 1460 mg/m³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 730 mg/m³ 8 Stunden	STEL: 1468 mg/m³ 15 minutach TWA: 734 mg/m³ 8 godzinach	TWA: 200 ppm 8 timer TWA: 734 mg/m³ 8 timer STEL: 400 ppm 15 minutter. value from the regulation STEL: 1468 mg/m³ 15 minutter. value from the regulation
Methanol	Haut MAK-KZGW: 800 ppm 15 Minuten MAK-KZGW: 1040 mg/m³ 15 Minuten	TWA: 200 ppm 8 timer TWA: 260 mg/m³ 8 timer STEL: 400 ppm 15 minutter STEL: 520 mg/m³ 15	Haut/Peau STEL: 400 ppm 15 Minuten STEL: 520 mg/m³ 15 Minuten	STEL: 300 mg/m³ 15 minutach TWA: 100 mg/m³ 8 godzinach	TWA: 100 ppm 8 timer TWA: 130 mg/m³ 8 timer STEL: 150 ppm 15 minutter. value calculated

	MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 260 mg/m^3 8 Stunden	minutter Hud	TWA: 200 ppm 8 Stunden TWA: 260 mg/m^3 8 Stunden		STEL: 162.5 mg/m^3 15 minutter. value calculated Hud
Methyl ethyl ketone	Haut MAK-KZGW: 200 ppm 15 Minuten MAK-KZGW: 590 mg/m^3 15 Minuten MAK-TMW: 100 ppm 8 Stunden MAK-TMW: 295 mg/m^3 8 Stunden	TWA: 50 ppm 8 timer TWA: 145 mg/m^3 8 timer STEL: 900 mg/m^3 15 minutter STEL: 300 ppm 15 minutter Hud	Haut/Peau STEL: 200 ppm 15 Minuten STEL: 590 mg/m^3 15 Minuten TWA: 200 ppm 8 Stunden TWA: 590 mg/m^3 8 Stunden	STEL: 900 mg/m^3 15 minutach TWA: 450 mg/m^3 8 godzinach	TWA: 75 ppm 8 timer TWA: 220 mg/m^3 8 timer STEL: 112.5 ppm 15 minutter. value calculated STEL: 275 mg/m^3 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Ethyl alcohol	TWA: 1000 mg/m^3	TWA-GVI: 1000 ppm 8 satima. TWA-GVI: 1900 mg/m^3 8 satima.	STEL: 1000 ppm 15 min		TWA: 1000 mg/m^3 8 hodinách. Ceiling: 3000 mg/m^3
Ethyl acetate	TWA: 734 mg/m^3 TWA: 200 ppm STEL : 1468 mg/m^3 STEL : 400 ppm	TWA-GVI: 200 ppm 8 satima. TWA-GVI: 734 mg/m^3 8 satima. STEL-KGVI: 400 ppm 15 minutama. STEL-KGVI: 1468 mg/m^3 15 minutama.	TWA: 734 mg/m^3 8 hr. TWA: 200 ppm 8 hr. STEL: 1468 mg/m^3 15 min STEL: 400 ppm 15 min	STEL: 1468 mg/m^3 STEL: 400 ppm TWA: 734 mg/m^3 TWA: 200 ppm	TWA: 700 mg/m^3 8 hodinách. Ceiling: 900 mg/m^3
Methanol	TWA: 200 ppm TWA: 260.0 mg/m^3 Skin notation	kože TWA-GVI: 200 ppm 8 satima. TWA-GVI: 260 mg/m^3 8 satima.	TWA: 200 ppm 8 hr. TWA: 260 mg/m^3 8 hr. STEL: 600 ppm 15 min STEL: 780 mg/m^3 15 min Skin	Skin-potential for cutaneous absorption TWA: 200 ppm TWA: 260 mg/m^3	TWA: 250 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 1000 mg/m^3
Methyl ethyl ketone	TWA: 590 mg/m^3 STEL : 885 mg/m^3	TWA-GVI: 200 ppm 8 satima. TWA-GVI: 600 mg/m^3 8 satima. STEL-KGVI: 300 ppm 15 minutama. STEL-KGVI: 900 mg/m^3 15 minutama.	TWA: 200 ppm 8 hr. TWA: 600 mg/m^3 8 hr. STEL: 300 ppm 15 min STEL: 900 mg/m^3 15 min Skin	STEL: 300 ppm STEL: 900 mg/m^3 TWA: 200 ppm TWA: 600 mg/m^3	TWA: 600 mg/m^3 8 hodinách. Ceiling: 900 mg/m^3

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Ethyl alcohol	TWA: 500 ppm 8 tundides. TWA: 1000 mg/m^3 8 tundides. STEL: 1000 ppm 15 minutites. STEL: 1900 mg/m^3 15 minutites.		TWA: 1000 ppm TWA: 1900 mg/m^3	STEL: 3800 mg/m^3 15 percekben. CK TWA: 1900 mg/m^3 8 órában. AK	TWA: 1000 ppm 8 klukkustundum. TWA: 1900 mg/m^3 8 klukkustundum. Ceiling: 2000 ppm Ceiling: 3800 mg/m^3
Ethyl acetate	TWA: 150 ppm 8 tundides. TWA: 500 mg/m^3 8 tundides. STEL: 300 ppm 15 minutites. STEL: 1100 mg/m^3 15 minutites.	TWA: 734 ppm 8 hr TWA: 200 mg/m^3 8 hr STEL: 1468 ppm 15 min STEL: 400 mg/m^3 15 min	STEL: 400 ppm STEL: 1468 mg/m^3 TWA: 200 ppm TWA: 734 mg/m^3	STEL: 1468 mg/m^3 15 percekben. CK TWA: 734 mg/m^3 8 órában. AK	TWA: 150 ppm 8 klukkustundum. TWA: 540 mg/m^3 8 klukkustundum. Ceiling: 300 ppm Ceiling: 1080 mg/m^3
Methanol	Nahk TWA: 200 ppm 8 tundides. TWA: 250 mg/m^3 8 tundides. STEL: 250 ppm 15 minutites. STEL: 350 mg/m^3 15 minutites.	Skin notation TWA: 200 ppm 8 hr TWA: 260 mg/m^3 8 hr	skin - potential for cutaneous absorption STEL: 250 ppm STEL: 325 mg/m^3 TWA: 200 ppm TWA: 260 mg/m^3	TWA: 260 mg/m^3 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 200 ppm 8 klukkustundum. TWA: 260 mg/m^3 8 klukkustundum. Skin notation Ceiling: 400 ppm Ceiling: 520 mg/m^3
Methyl ethyl ketone	TWA: 200 ppm 8	TWA: 200 ppm 8 hr	STEL: 300 ppm	STEL: 900 mg/m^3 15	STEL: 300 ppm

	tundides. TWA: 600 mg/m^3 8 tundides. STEL: 300 ppm 15 minutites. STEL: 900 mg/m^3 15 minutites.	TWA: 600 mg/m^3 8 hr STEL: 300 ppm 15 min STEL: 900 mg/m^3 15 min	STEL: 900 mg/m^3 TWA: 200 ppm TWA: 600 mg/m^3	percekben. CK TWA: 600 mg/m^3 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 900 mg/m^3 TWA: 50 ppm 8 klukkustundum. TWA: 145 mg/m^3 8 klukkustundum. Skin notation

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Ethyl alcohol	TWA: 1000 mg/m^3	TWA: 500 ppm IPRD TWA: 1000 mg/m^3 IPRD STEL: 1000 ppm STEL: 1900 mg/m^3			TWA: 1000 ppm 8 ore TWA: 1900 mg/m^3 8 ore STEL: 5000 ppm 15 minute STEL: 9500 mg/m^3 15 minute
Ethyl acetate	STEL: 1468 mg/m^3 STEL: 400 ppm TWA: 200 mg/m^3 TWA: 54 ppm	Ceiling: 300 ppm Ceiling: 1100 mg/m^3 TWA: 150 ppm IPRD TWA: 500 mg/m^3 IPRD	TWA: 734 mg/m^3 8 Stunden TWA: 200 ppm 8 Stunden STEL: 1468 mg/m^3 15 Minuten STEL: 400 ppm 15 Minuten	TWA: 200 ppm TWA: 734 mg/m^3 STEL: 400 ppm 15 minuti STEL: 1468 mg/m^3 15 minuti	TWA: 111 ppm 8 ore TWA: 400 mg/m^3 8 ore STEL: 139 ppm 15 minute STEL: 500 mg/m^3 15 minute
Methanol	skin - potential for cutaneous exposure TWA: 200 ppm TWA: 260 mg/m^3	TWA: 200 ppm IPRD TWA: 260 mg/m^3 IPRD Oda	Possibility of significant uptake through the skin TWA: 200 ppm 8 Stunden TWA: 260 mg/m^3 8 Stunden	possibility of significant uptake through the skin TWA: 200 ppm TWA: 260 mg/m^3	Skin notation TWA: 200 ppm 8 ore TWA: 260 mg/m^3 8 ore
Methyl ethyl ketone	STEL: 300 ppm STEL: 900 mg/m^3 TWA: 67 ppm TWA: 200 mg/m^3		TWA: 200 ppm 8 Stunden TWA: 600 mg/m^3 8 Stunden STEL: 300 ppm 15 Minuten STEL: 900 mg/m^3 15 Minuten	TWA: 200 ppm TWA: 600 mg/m^3 STEL: 300 ppm 15 minuti STEL: 900 mg/m^3 15 minuti	TWA: 200 ppm 8 ore TWA: 600 mg/m^3 8 ore STEL: 300 ppm 15 minute STEL: 900 mg/m^3 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Ethyl alcohol	TWA: 1000 mg/m^3 2391 MAC: 2000 mg/m^3	Ceiling: 1920 mg/m^3 TWA: 500 ppm TWA: 960 mg/m^3	TWA: 960 mg/m^3 8 urah TWA: 500 ppm 8 urah STEL: 1000 ppm 15 minutah STEL: 1920 mg/m^3 15 minutah	Indicative STEL: 1000 ppm 15 minuter Indicative STEL: 1900 mg/m^3 15 minuter TLV: 500 ppm 8 timmar. NGV TLV: 1000 mg/m^3 8 timmar. NGV	
Ethanedioic acid, dihydrate	Skin notation MAC: 1 mg/m^3			Indicative STEL: 2 mg/m^3 15 minuter TLV: 1 mg/m^3 8 timmar. NGV	
Ethyl acetate	TWA: 50 mg/m^3 2417 MAC: 200 mg/m^3	Ceiling: 1100 mg/m^3 TWA: 200 ppm TWA: 734 mg/m^3	TWA: 200 ppm 8 urah TWA: 734 mg/m^3 8 urah STEL: 400 ppm 15 minutah STEL: 1468 mg/m^3 15 minutah	Binding STEL: 300 ppm 15 minuter Binding STEL: 1100 mg/m^3 15 minuter TLV: 150 ppm 8 timmar. NGV TLV: 550 mg/m^3 8 timmar. NGV	
Methanol	TWA: 5 mg/m^3 1250 Skin notation MAC: 15 mg/m^3	Potential for cutaneous absorption TWA: 200 ppm TWA: 260 mg/m^3	TWA: 200 ppm 8 urah TWA: 260 mg/m^3 8 urah Koža STEL: 800 ppm 15 minutah STEL: 1040 mg/m^3 15 minutah	Indicative STEL: 250 ppm 15 minuter Indicative STEL: 350 mg/m^3 15 minuter TLV: 200 ppm 8 timmar. NGV TLV: 250 mg/m^3 8 timmar. NGV Hud	Deri TWA: 200 ppm 8 saat TWA: 260 mg/m^3 8 saat
Methyl ethyl ketone	TWA: 200 mg/m^3 0421	Ceiling: 900 mg/m^3	TWA: 200 ppm 8 urah	Binding STEL: 300 ppm	TWA: 200 ppm 8 saat

	MAC: 400 mg/m^3	TWA: 200 ppm TWA: 600 mg/m^3	TWA: 600 mg/m^3 8 urah Koža STEL: 300 ppm 15 minutah STEL: 900 mg/m^3 15 minutah	15 minuter Binding STEL: 900 mg/m^3 15 minuter TLV: 50 ppm 8 timmar. NGV TLV: 150 mg/m^3 8 timmar. NGV	TWA: 600 mg/m^3 8 saat STEL: 300 ppm 15 dakika STEL: 900 mg/m^3 15 dakika

Biological limit values
List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

Component	European Union	United Kingdom	France	Spain	Germany
Methanol			Methanol: 15 mg/L urine end of shift	Methanol: 15 mg/L urine end of shift	Methanol: 15 mg/L urine (end of shift) Methanol: 15 mg/L urine (for long-term exposures: at the end of the shift after several shifts)
Methyl ethyl ketone		Butan-2-one: 70 µmol/L urine post shift	Methylethylketone: 2 mg/L urine end of shift	Methyl ethyl ketone: 2 mg/L urine end of shift	2-Butanone: 2 mg/L urine (end of shift)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Methanol					Methanol: 6 mg/L urine end of shift
Methyl ethyl ketone					Methylethylketone: 2 mg/L urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Methanol			Methanol: 30 mg/L urine end of exposure or work shift Methanol: 30 mg/L urine after all work shifts for long-term exposure		

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / **Derived Minimum Effect Level (DMEL)**
See table for values

Component	Acute effects local (Oral)	Acute effects systemic (Oral)	Chronic effects local (Oral)	Chronic effects systemic (Oral)
Ethyl alcohol 64-17-5 (74.15)		DNEL = 87 mg/kg bw/d		

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Ethyl alcohol 64-17-5 (74.15)				DNEL = 343mg/kg bw/day
Ethyl acetate 141-78-6 (8.88)				DNEL = 63mg/kg bw/day
Methanol 67-56-1 (4.44)		DNEL = 20mg/kg bw/day		DNEL = 20mg/kg bw/day

Methyl ethyl ketone 78-93-3 (1.33)				DNEL = 1161mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Ethyl alcohol 64-17-5 (74.15)	DNEL = 1900mg/m³			DNEL = 950mg/m³
Ethyl acetate 141-78-6 (8.88)	DNEL = 1468 mg/m³ 400 ppm	DNEL = 1468 mg/m³ 400 ppm	DNEL = 734 mg/m³ 200 ppm	DNEL = 734mg/m³
Methanol 67-56-1 (4.44)	DNEL = 130mg/m³	DNEL = 130mg/m³	DNEL = 130mg/m³	DNEL = 130mg/m³
Methyl ethyl ketone 78-93-3 (1.33)				DNEL = 600mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Ethyl acetate 141-78-6 (8.88)	PNEC = 0.24mg/L	PNEC = 1.15mg/kg sediment dw	PNEC = 1.65mg/L	PNEC = 650mg/L	PNEC = 0.148mg/kg soil dw
Methanol 67-56-1 (4.44)	PNEC = 20.8mg/L	PNEC = 77mg/kg sediment dw	PNEC = 1540mg/L	PNEC = 100mg/L	PNEC = 100mg/kg soil dw
Methyl ethyl ketone 78-93-3 (1.33)	PNEC = 55.8mg/L	PNEC = 284.74mg/kg sediment dw	PNEC = 55.8mg/L	PNEC = 709mg/L	PNEC = 22.5mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Ethyl acetate 141-78-6 (8.88)	PNEC = 0.024mg/L	PNEC = 0.115mg/kg sediment dw		PNEC = 0.2g/kg food	
Methanol 67-56-1 (4.44)	PNEC = 2.08mg/L	PNEC = 7.7mg/kg sediment dw			
Methyl ethyl ketone 78-93-3 (1.33)	PNEC = 55.8mg/L	PNEC = 284.7mg/kg sediment dw		PNEC = 1000mg/kg food	

8.2. Exposure controls

Engineering Measures
Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	Alcohol	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Highly flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	14 °C / 57.2 °F	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Ethyl alcohol	-0.32	
Ethyl acetate	0.73	
Methanol	-0.74	
Methyl ethyl ketone	0.29	
Vapor Pressure	<=1100 hPa @ 50 °C	
Density / Specific Gravity	No data available	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Explosive Properties Vapors may form explosive mixtures with air

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Oxidizing agent.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ethyl alcohol	LD50 = 10470 mg/kg OECD 401 (Rat) 3450 mg/kg (Mouse)	-	LC50 = 117-125 mg/l (4h) OECD 403 (rat) 20000 ppm/10H (rat)
Ethanedioic acid, dihydrate	LD50 = 375 mg/kg (Rat)	-	-
Ethyl acetate	10,200 mg/kg (Rat)	> 20 mL/kg (Rabbit) > 18000 mg/kg (Rabbit)	58 mg/l (rat; 8 h)
Methanol	LD50 = 1187 – 2769 mg/kg (Rat)	LD50 = 17100 mg/kg (Rabbit)	LC50 = 128.2 mg/L (Rat) 4 h
Methyl ethyl ketone	LD50 = 2483 mg/kg (Rat)	LD50 = 5000 mg/kg (Rabbit)	LC50 = 11700 ppm (Rat) 4 h

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

Component	Test method	Test species	Study result
Ethyl alcohol 64-17-5 (74.15)	Mouse Ear Swelling Test (MEST) -- OECD Test Guideline 429 Local Lymph Node Assay	mouse -- mouse	non-sensitising -- non-sensitising
Ethyl acetate 141-78-6 (8.88)	OECD Test Guideline 406	guinea pig	- non-sensitising
Methanol 67-56-1 (4.44)	OECD Test Guideline 406 Guinea Pig Maximisation Test (GPMT)	guinea pig	non-sensitising

(e) germ cell mutagenicity; No data available

Component	Test method	Test species	Study result
Ethyl alcohol 64-17-5 (74.15)	AMES test OECD Test Guideline 471 -- Gene cell mutation OECD Test Guideline 476	in vitro Bacteria -- in vitro Mammalian	negative -- negative
Ethyl acetate 141-78-6 (8.88)	OECD Test Guideline 471 AMES test -- OECD Test Guideline 473 Chromosomal aberration assay -- OECD Test Guideline 476 Gene cell mutation -- OECD Test Guideline 474 Mouse micronucleus assay	in vitro Bacteria -- in vitro Mammalian -- in vitro Mammalian -- in vivo Mammalian	negative -- negative -- negative -- negative

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; No data available

Component	Test method	Test species / Duration	Study result
Ethyl alcohol 64-17-5 (74.15)	OECD Test Guideline 416 -- OECD Test Guideline 414	Oral / mouse 2 Generation -- Inhalation / Rat	NOAEL = 13.8 g/kg/day -- NOAEC = 16000 ppm
Ethyl acetate 141-78-6 (8.88)	OECD Test Guideline 416 -- OECD Test Guideline 414	Oral mouse 2 Generation -- Inhalation Rat	NOAEL = 26400 mg/kg bw/day -- NOAEC = 73300 mg/m³
Methanol 67-56-1 (4.44)	OECD Test Guideline 416	Rat / Inhalation 2 Generation	NOAEC = 1.3 mg/l (air)

(h) STOT-single exposure; Category 2

Results / Target organs Central nervous system (CNS), Optic nerve.

(i) STOT-repeated exposure; Category 2

Target Organs Kidney.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties .
Assess endocrine disrupting properties for human health Contains a substance on the National Authorities Endocrine Disruptor Lists

Component	EU National Authorities Endocrine Disruptor Lists - Health
Methyl ethyl ketone 78-93-3 (1.33)	List II

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Contains a substance which is:. Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Ethyl alcohol	Fathead minnow (Pimephales promelas) LC50 = 14200 mg/l/96h	EC50 = 9268 mg/L/48h EC50 = 10800 mg/L/24h	EC50 (72h) = 275 mg/l (Chlorella vulgaris)
Ethanedioic acid, dihydrate	LC50 = 160 mg/L/48h (Carassius auratus)		
Ethyl acetate	Fathead minnow: LC50: 230 mg/l/ 96h Gold orfe: LC50: 270 mg/L/48h	EC50 = 717 mg/L/48h	EC50 = 3300 mg/L/48h
Methanol	Pimephales promelas: LC50 > 10000 mg/L 96h	EC50 > 10000 mg/L 24h	
Methyl ethyl ketone	Lepomis macrochirus: LC50=3,22 g/L 96 h	EC50: = 5091 mg/L, 48h (Daphnia magna) EC50: 4025 - 6440 mg/L, 48h Static (Daphnia magna) EC50: > 520 mg/L, 48h (Daphnia magna)	

Component	Microtox	M-Factor
Ethyl alcohol	Photobacterium phosphoreum:EC50 = 34634 mg/L/30 min Photobacterium phosphoreum:EC50 = 35470 mg/L/5 min	
Ethyl acetate	EC50 = 1180 mg/L 5 min EC50 = 1500 mg/L 15 min EC50 = 5870 mg/L 15 min EC50 = 7400 mg/L 2 h	
Methanol	EC50 = 39000 mg/L 25 min EC50 = 40000 mg/L 15 min EC50 = 43000 mg/L 5 min	
Methyl ethyl ketone	EC50 = 3403 mg/L 30 min EC50 = 3426 mg/L 5 min	

12.2. Persistence and degradability

Persistence Miscible with water, Persistence is unlikely, based on information available.

Component	Degradability

Ethyl alcohol 64-17-5 (74.15)	OECD 301E = 94%
Ethyl acetate 141-78-6 (8.88)	79 % (20 d) (OECD 301 D)
Methanol 67-56-1 (4.44)	DT50 ~ 17.2d >94% after 20d
Methyl ethyl ketone 78-93-3 (1.33)	98% (28d)

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Ethyl alcohol	-0.32	No data available
Ethyl acetate	0.73	30 dimensionless
Methanol	-0.74	<10 dimensionless
Methyl ethyl ketone	0.29	No data available

12.4. Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN1992
14.2. UN proper shipping name Flammable liquid, toxic, n.o.s.
Technical Shipping Name (METHANOL, Oxalic acid dihydrate)
14.3. Transport hazard class(es) 3
Subsidiary Hazard Class 6.1
14.4. Packing group III

ADR

14.1. UN number UN1992
14.2. UN proper shipping name Flammable liquid, toxic, n.o.s.
Technical Shipping Name (METHANOL, Oxalic acid dihydrate)
14.3. Transport hazard class(es) 3
Subsidiary Hazard Class 6.1
14.4. Packing group III

IATA

14.1. UN number UN1992
14.2. UN proper shipping name Flammable liquid, toxic, n.o.s.
Technical Shipping Name (METHANOL, Oxalic acid dihydrate)
14.3. Transport hazard class(es) 3
Subsidiary Hazard Class 6.1
14.4. Packing group III

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Ethyl alcohol	64-17-5	200-578-6	-	-	X	X	KE-13217	X	X
Ethanedioic acid, dihydrate	6153-56-6	-	-	-	X	X	-	X	X
Ethyl acetate	141-78-6	205-500-4	-	-	X	X	KE-00047	X	X
Methanol	67-56-1	200-659-6	-	-	X	X	KE-23193	X	X
Methyl ethyl ketone	78-93-3	201-159-0	-	-	X	X	KE-24094	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Ethyl alcohol	64-17-5	X	ACTIVE	X	-	X	X	X
Ethanedioic acid, dihydrate	6153-56-6	-	-	-	-	X	X	X
Ethyl acetate	141-78-6	X	ACTIVE	X	-	X	X	X
Methanol	67-56-1	X	ACTIVE	X	-	X	X	X
Methyl ethyl ketone	78-93-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Ethyl alcohol	64-17-5	-	-	-
Ethanedioic acid, dihydrate	6153-56-6	-	-	-
Ethyl acetate	141-78-6	-	Use restricted. See item 75. (see link for restriction details)	-
Methanol	67-56-1	-	Use restricted. See item 69. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	-
Methyl ethyl ketone	78-93-3	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Ethyl alcohol	64-17-5	Not applicable	Not applicable
Ethanedioic acid, dihydrate	6153-56-6	Not applicable	Not applicable
Ethyl acetate	141-78-6	Not applicable	Not applicable
Methanol	67-56-1	500 tonne	5000 tonne
Methyl ethyl ketone	78-93-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class

Ethyl alcohol	WGK1	
Ethyl acetate	WGK1	
Methanol	WGK 2	Class I : 20 mg/m³ (Massenkonzentration)
Methyl ethyl ketone	WGK1	

Component	France - INRS (Tables of occupational diseases)
Ethyl alcohol	Tableaux des maladies professionnelles (TMP) - RG 84
Ethyl acetate	Tableaux des maladies professionnelles (TMP) - RG 84
Methanol	Tableaux des maladies professionnelles (TMP) - RG 84
Methyl ethyl ketone	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Ethyl alcohol 64-17-5 (74.15)		Group I	
Ethyl acetate 141-78-6 (8.88)		Group I	
Methanol 67-56-1 (4.44)	Prohibited and Restricted Substances	Group I	
Methyl ethyl ketone 78-93-3 (1.33)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H318 - Causes serious eye damage
H371 - May cause damage to organs
H373 - May cause damage to organs through prolonged or repeated exposure
EUH066 - Repeated exposure may cause skin dryness or cracking
H225 - Highly flammable liquid and vapor
H301 - Toxic if swallowed
H302 - Harmful if swallowed
H311 - Toxic in contact with skin
H312 - Harmful in contact with skin
H319 - Causes serious eye irritation
H331 - Toxic if inhaled
H336 - May cause drowsiness or dizziness
H370 - Causes damage to organs

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.
Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Prepared By	Health, Safety and Environmental Department
Revision Date	19-Mar-2024
Revision Summary	New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

End of Safety Data Sheet